Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti |

AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

ML Engineer specializing in **LLM evaluation pipelines and AI observability** - built AgentProbe, an open-source eval toolkit with LLM-powered judges, 5-metric eval engine, trace enrichment, and automated fix agents (LangChain, LangGraph, ChromaDB, FastAPI). 8+ years backend engineering at scale (1M+ records, 100K+ MAU, Python/AWS). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Demo: *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

LLM Evaluation Pipelines | LLM-Powered Judges | AI Agent Observability | Trace Enrichment & Dashboards | Regression Prevention | LangChain / LangGraph | RAG Pipelines / ChromaDB | Python (Backend) | FastAPI / REST APIs | AWS Distributed Pipelines | PostgreSQL / SQL Tuning | Go (Ramping)

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built **distributed data pipelines** on AWS (S3, Glue, EC2) processing **1M+ records** with automated quality checks, Libsodium encryption, and zero security incidents - eliminating 80% of manual operational workload via Python and PL-SQL automation.
- Improved production query performance by **30%** by architecting PostgreSQL schemas, data access layers, and indexing strategies across 20+ projects - demonstrating the analytical rigor needed to connect offline metrics to real system improvements.
- Reduced deployment times by **40%** and maintained sub-10 bug count for 6+ consecutive months by building 20+ full-stack modules (Python, PostgreSQL) with TDD/BDD gates across 120+ Agile sprints - 100K+ monthly active users.
- Cut microservices execution time by **50%** by leading Docker-based containerization for 30%+ of modules, delivering observable, scalable backend architecture supporting multi-tenant production environments.
- Improved system stability by **35%**, resolved 100+ high-priority bugs, and maintained **99.5% uptime** - building the quality-gate mindset that prevents regressions before they reach users.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Delivered targeted Python and database mentoring for 30+ undergraduates through sprint-style feedback cycles.

PROJECTS

AgentProbe: LLM Evaluation & Observability Toolkit OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · Ollama · React · Azure

Built a **large-scale LLM evaluation system** with LLM-powered judges scoring correctness, completeness, retrieval quality, hallucination rate, and compliance across agent runs. Architecture: 3 LangGraph agents (Orchestrator, **RAG Tracer** for trace enrichment, **Auto Fix** for automated prompt iteration), smart LLM Router for dynamic tool routing, **15-endpoint FastAPI backend**, React observability dashboard with per-query trace visualization. Supports Ollama, OpenAI, Anthropic, Groq - runs 100% locally at zero cost. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · NumPy · Pandas

Built NLP pipeline mapping natural language queries (EN/ES) to FHIR APIs via LLMs, BERT, and embeddings. HIPAA-compliant RBAC, containerized. **Reduced retrieval time ~60%**. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: UK Road Accident Analysis

Python · R · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

Predictive modeling on 100K+ accident records. **85%+ accuracy**; interactive GitHub Pages dashboard for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS; Go (ramping)

AI & Eval: LangChain, LangGraph, LLM Judges, ChromaDB, RAG, Embeddings, Ollama, OpenAI, Anthropic, Groq, FastAPI, Prompt Engineering, Agentic AI

Databases: PostgreSQL, MySQL, Oracle, NoSQL, GraphQL - schema design, query tuning, indexing

Cloud & DevOps: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), RabbitMQ, Microservices

ML & NLP: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, BERT, spaCy, Hugging Face, Feature Engineering, Predictive Modeling

Testing: TDD/BDD, Integration Testing, Selenium, Git, JIRA, Postman, Agile/Scrum